

Binary cross entropy is used with sigmoid activation functions

$$L = - [y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$$

$$\hat{y} = \sigma(z)$$

$$z = wx + b$$

differentiating binary cross entropy eqn:

$$\frac{\partial L}{\partial \hat{y}} = -\frac{y}{\hat{y}} + \frac{1-y}{(1-\hat{y})} = \frac{\hat{y}-y}{\hat{y}(1-\hat{y})}$$

differentiating sigmoid eqn:

$$\hat{y} = \sigma(z)$$

$$\text{and, } \frac{\partial \hat{y}}{\partial z} = \hat{y}(1-\hat{y})$$

applying chain rule:

$$\frac{\partial L}{\partial z} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z}$$

$$(1-\hat{y}) \times (1-\hat{y}) = \frac{\hat{y}-y}{\hat{y}(1-\hat{y})} \cdot \hat{y}(1-\hat{y})$$
$$= \hat{y}-y$$

this result is used in backpropagation;

again,

$$\frac{\partial z}{\partial w} = X \text{ , and } \frac{\partial z}{\partial b} = 1$$

now for weight,

$$\begin{aligned} \frac{\partial L}{\partial w} &= \frac{\partial L}{\partial z} \cdot \frac{\partial z}{\partial w} \\ &= (\hat{y} - y) X \end{aligned}$$

now, for bias,

$$\begin{aligned} \frac{\partial L}{\partial b} &= \frac{\partial L}{\partial z} \times \frac{\partial z}{\partial b} \\ &= (\hat{y} - y) \cdot 1 \\ &= \hat{y} - y \end{aligned}$$

now, updated weights and bias is,

$$w_{\text{new}} = w - \eta \frac{\partial L}{\partial w} = w - \eta X (\hat{y} - y)$$

$$b_{\text{new}} = b - \eta \frac{\partial L}{\partial b} = b - \eta (\hat{y} - y)$$